EXP 14	**Exploring pro-code extension of a Zoho Creator application using Zoho Catalyst / custom functions**

Aim:

To extend the **Event Registration System** with pro-code features using custom functions in Zoho Creator and to understand how Zoho Catalyst can be used for advanced logic.

Theory:

Custom functions are reusable blocks of code in Zoho Creator. They can be written in **Deluge**, **NodeJs** or **Java**, and can be called from workflows, schedules, buttons or APIs. They help to add logic that drag-and-drop tools cannot give.

Zoho Catalyst is Zoho's serverless pro-code platform. It gives functions, a data store, file storage, cron jobs and APIs. A Creator app can call a Catalyst function through a URL or a connection when heavy or complex processing is needed, for example generating QR-code tickets, payment processing, or bulk reports.

Procedure:

1. Open the Event Registration System in Edit mode.
2. Go to **Workflow** > **Functions**.
3. Click **Create Function** and check the language options: Deluge, NodeJs and Java.
4. Enter the name **seatsLeftReport**, choose Deluge, namespace Default and return type void.
5. Write the code that finds the seats left for each event.
6. Save the function.
7. Execute the function to view the output in the log.
8. Study how the same logic can be moved to a Zoho Catalyst function for large data.

Program:

```
void seatsLeftReport()
{
  for each ev in Events[ID != 0]
  {
    used = 0;
     for each r in Event_Registration_Form[Event == ev.ID && Registration_Status == "Confirmed"]
    {
```

```
      used = used + r.Number_of_Participants;
    }
    info ev.Event_Name + " | Max: " + ev.Maximum_Participants + " | Confirmed: " +
used
        + " | Seats left: " + (ev.Maximum_Participants - used);
  }
}
```

Output:

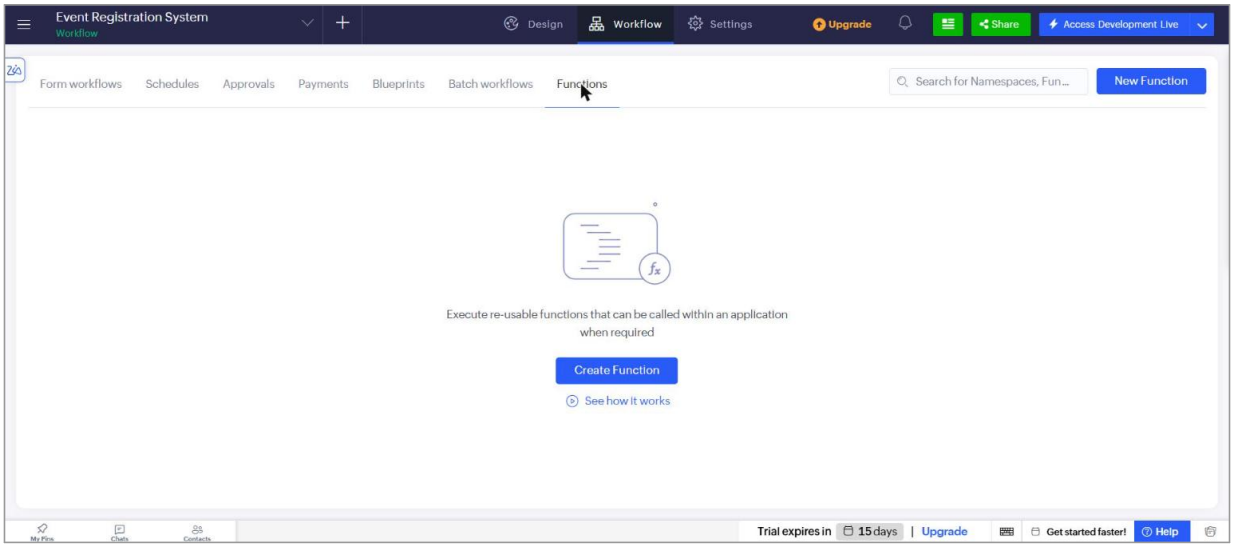


Fig. 14.1: Functions Section in the Workflow Builder

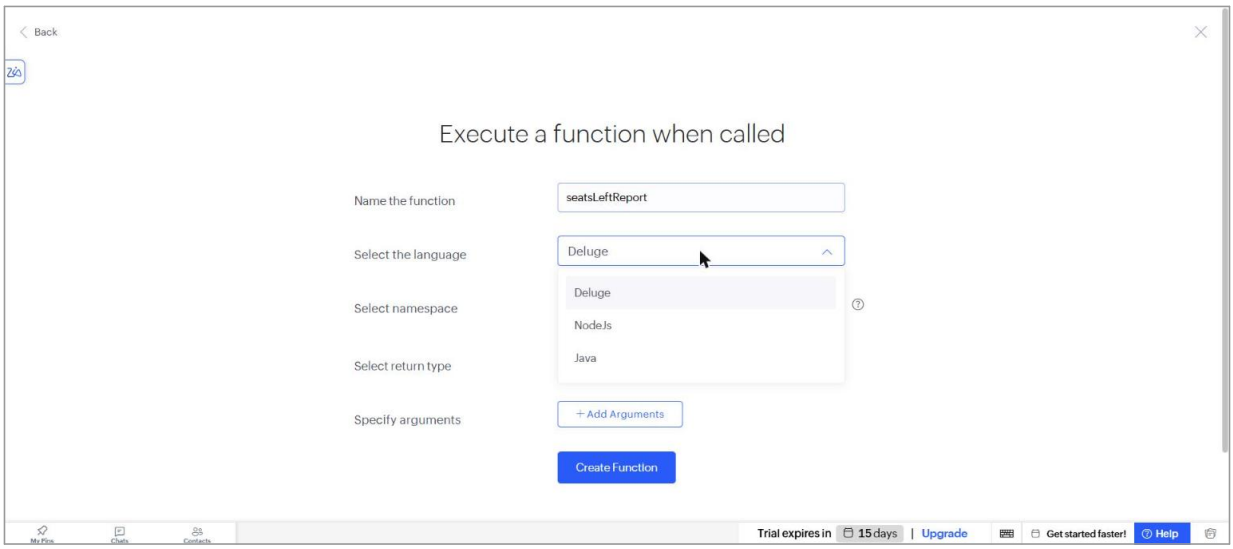


Fig. 14.2: Language Options - Deluge, NodeJs and Java

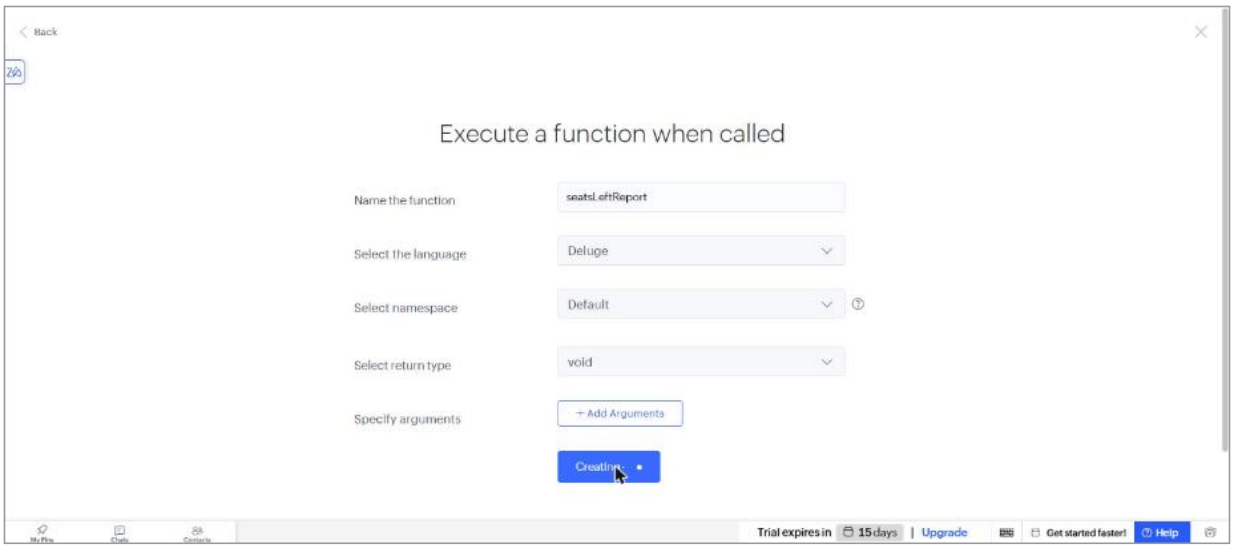


Fig. 14.3: Creating the seatsLeftReport Function

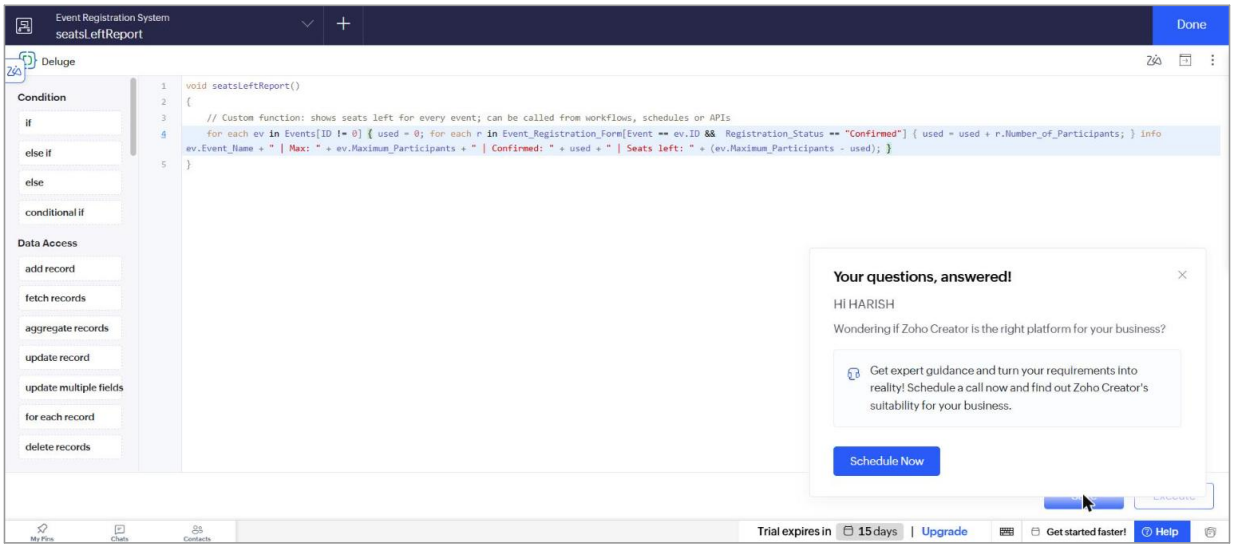


Fig. 14.4: Custom Function Code Saved in the Editor

Expected output of the function (based on the data in the app):

Event	Max	Confirmed	Seats Left
Hackathon 2026	10	5	5
AI Workshop	3	3	0
Tech Quiz	20	3	17

Zoho Catalyst Extension:

- Create a Catalyst project and add a **Basic I/O** or **Advanced I/O** function in NodeJs or Java.
- Move heavy logic (for example QR-code tickets or bulk seat reports) into the Catalyst function.

- Call the Catalyst function from a Creator workflow using invokeurl with a connection.
- Return the result to Creator and store it in a field or show it on the dashboard.

Result:

The Event Registration System was extended using a custom function in Zoho Creator, and the use of Zoho Catalyst for advanced pro-code logic was explored.